

5.4.2 Identify Security Threats

خطرے کی جلد پہچان نقصان سے پہلے اقدام ممکن بناتی ہے۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 11 of 15

5.4.2 Identify Security Threats

LEARN IT

Identifying security threats means **noticing signs** that data or systems may be at risk. Early identification helps you act before serious damage occurs.

Sign / indicator	What it likely means
Slow performance, missing files, or random pop-ups.	Malware or background unauthorised activity.
Failed login attempts or strange login locations.	Attempted unauthorised access .
Emails urging quick action, asking for an OTP, or containing bad grammar / strange links.	Classic phishing .
Unexpected deletion, copying or movement of records.	Data theft or insider misuse.
Sudden massive spike in server requests or a very slow website.	An ongoing DoS attack.

The first line of defence is also simple behaviour: avoid **unsafe downloads, cracked software, unverified websites and infected USB drives**.

WORKED EXAMPLE

Worked example: read the signs in a college computer lab in Khairpur.

1. Three PCs show pop-ups and Word files have vanished after students used a "game" USB. → Treat as **malware**. Disconnect from the network; run antivirus; stop USB use.
2. Google Classroom emails: "Your account will be deleted in 30 minutes. Confirm password here." Link spelling is goOgle. → **Phishing**. Do not click; report to the teacher.
3. The e-portal that was fine at 8:00 am is unreachable at 8:05; the technician sees a sudden spike in requests. → Suspect **DoS**.
4. Nadia's Gmail security page shows a login from another country at 4 a.m. → **Unauthorised access** attempt. Change password; turn on MFA.

REMEMBER IT

Slow/pop-ups → **malware**. **Odd logins** → **unauthorised access**. **Urgent OTP email** → **phishing**. **Files moving** → **theft**. **Traffic spike** → **DoS**.

COMMON MISTAKE

Mistake: Ignoring early signs ("the PC is just old") until the ransomware note appears.

Fix: One unusual login or one OTP email is enough to investigate. Early action is cheaper than recovery.

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CHECK YOURSELF

1. Random pop-ups and missing files most likely indicate what?
2. An email with bad grammar that demands your OTP is a sign of what?
3. Name four unsafe habits that invite these threats.

Answers: 1. Malware (or background unauthorised activity). 2. Phishing. 3. Unsafe downloads, cracked software, unverified websites, infected USB drives.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An IDE is:

- a. One app to write, run and debug code
- b. A printer
- c. A sort
- d. A cable

MCQ 2. VS Code is:

- a. A popular editor/IDE
- b. A DBMS
- c. An OSI layer
- d. A gate

MCQ 3. The terminal/run button is for:

- a. Executing the program
- b. Drawing pie charts only
- c. Sending TCP SYN only
- d. K-maps

MCQ 4. Syntax highlighting helps you:

- a. See keywords and mistakes
- b. Sort faster
- c. Encrypt disks
- d. Route packets

MCQ 5. A file explorer in VS Code shows:

- a. Project files
- b. OSI layer 1
- c. NAND groups
- d. Beachhead users

MCQ 6. Debugging in an IDE uses:

- a. Breakpoints and variable watch
- b. Only bubble sort
- c. Only pie charts
- d. Only Agile stickers

Extra short questions

1. Define IDE.
2. Name four VS Code areas.
3. Why not use Notepad only?
4. What is an extension?

Extra long question

1. Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Integrated Development Environment: editor + run + debug + files in one place.
2. Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
3. No easy run, no highlighting, no debugger.
4. An add-on (Python plugin) that teaches VS Code a language.

Long

1. New file .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.